Lagrange Method for PDEs

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Lagrange's method solving Linear Partial Differential Equations

Solve the following PDEs :

1. $(-\boldsymbol{a}+\boldsymbol{x})\boldsymbol{p}+(-\boldsymbol{b}+\boldsymbol{y})\boldsymbol{q}=(-\boldsymbol{c}+\boldsymbol{z})$

Solution : Given PDE is

$(-a+x)p+(-b+y)q=(-c+z)$ .... (i)

The Lagrange's auxiliary equations for (i) are

$\frac{dx}{x-a}=\frac{dy}{y-b}=\frac{dz}{z-c}$ .... (ii)

From 1st and 2nd fractions, we get

$$\frac{dx}{x-a}=\frac{dy}{y-b}$$

$\Rightarrow \int\frac{dx}{x-a}=\int\frac{dy}{y-b}+logc_1$ where $logc_1$ is an integrating constant

$$\Rightarrow \log(x-a)=\log(y-b)+logc_1$$

$$\Rightarrow \log(x-a)-\log(y-b)= logc_1$$

$$\Rightarrow logc_1=log\frac{x-a}{y-b}$$

$$\therefore c_1=\frac{x-a}{y-b}$$

From 1st and 3rd fractions, we get

$dx \qquad dz$

From 1st and 3rd fractions, we get

$$\frac{dx}{x-a} = \frac{dz}{z-c}$$

$\Rightarrow \int \frac{dx}{x-a} = \int \frac{dz}{z-c} + logc_2$ where $logc_2$ is an integrating constant

$$\Rightarrow \log(x-a) = \log(z-c) + logc_2$$

$$\Rightarrow \log(x-a) - \log(z-c) = logc_2$$

$$\Rightarrow logc_2 = log\frac{x-a}{z-c}$$

$$\therefore c_2 = \frac{x-a}{z-c}$$

$\therefore$ *The required general solution will be*

$$\varphi(c_1, c_2) = 0,$$

$i.e., \varphi\left(\frac{x-a}{y-b}, \frac{x-a}{z-c}\right) = 0,$ φ *is an arbitrary function.* **Answer**

2. $xp + yq = z$

Solution : Given PDE is,

$xp + yq = z$ .... (i)

The Lagrange's auxiliary equations for (i) are

$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$ .... (ii)

From 1st and 2nd fractions, we get

$$\frac{dx}{x} = \frac{dy}{y}$$

$\Rightarrow \int \frac{dx}{x} = \int \frac{dy}{y} + \ logc_1$ where $logc_1$is an integrating constant.

$\Rightarrow logx = logy + logc_1$

$\Rightarrow logx - logy = \ logc_1$

$\Rightarrow logc_1 = log\frac{x}{y}$

$\Rightarrow c_1 = \frac{x}{y}$ .... (iii)

$$\frac{dx}{x} = \frac{dz}{z}$$

$\Rightarrow \int \frac{dx}{x} = \int \frac{dz}{z} + \ logc_2$ where $logc_2$is an integrating constant.

$\Rightarrow logx = logz + logc_2$

$\Rightarrow logx - logz = \ logc_2$

$\Rightarrow logc_2 = log\frac{x}{z}$

$\Rightarrow c_2 = \frac{x}{z}$ .... (iv)

$\therefore The\ required\ general\ solution\ will\ be$

$\varphi(c_1, c_2) = 0,$

$i.e., \varphi\left(\frac{x}{y}, \frac{x}{z}\right) = 0,\ \varphi\ is\ an\ arbitrary\ function.$ **Answer**

$$\frac{dx}{x} = \frac{dz}{z}$$

$\Rightarrow \int \frac{dx}{x} = \int \frac{dz}{z} + logc_2$ where $logc_2$is an integrating constant.

$$\Rightarrow logx = logz + logc_2$$

$$\Rightarrow logx - logz = logc_2$$

$$\Rightarrow logc_2 = log\frac{x}{z}$$

$\Rightarrow c_2 = \frac{x}{z}$ .... (iv)

$\therefore$ *The required general solution will be*

$$\varphi(c_1, c_2) = 0,$$

$i.e., \varphi\left(\frac{x}{y}, \frac{x}{z}\right) = 0,\ \varphi$ *is an arbitrary function.* **Answer**

3. $\boldsymbol{p + q = 1}$

$i.e., \varphi(x - y, y - z) = 0,\ \varphi$ *is an arbitrary function.* **Answer**

4. $\boldsymbol{x^2p + y^2q = z^2}$

$i.e., \varphi\left(\frac{1}{y} - \frac{1}{x}, \frac{1}{z} - \frac{1}{y}\right) = 0,\ \varphi$ *is an arbitrary function.* **Answer**

6. $\boldsymbol{\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = sinx}$

7. $\boldsymbol{yzp + 2xq = xy}$

Solution : Given PDE is,

$yzp + 2xq = xy$ .... (i)

7. $\boldsymbol{yzp + 2xq = xy}$

Solution : Given PDE is,

$yzp + 2xq = xy$ ……………………… (i)

The Lagrange's auxiliary equations for (i) are

$\frac{dx}{yz} = \frac{dy}{2x} = \frac{dz}{xy}$ ……………………… (ii)

From 1st and 3rd fractions of (ii), we get

$$\frac{dx}{yz} = \frac{dz}{xy}$$

$i.e., \varphi(z^2 - x^2, 4z - y^2) = 0,\ \varphi\ is\ an\ arbitrary\ function.$ **Answer**

9. $\boldsymbol{yzp + zxq = xy}$

10. $\boldsymbol{zp = x}$

$i.e., \varphi(z^2 - x^2, y) = 0,\ \varphi\ is\ an\ arbitrary\ function.$ **Answer**

11. $\boldsymbol{y^2p + x^2q = x^2y^2z^2}$